Shashank Mahesh

Santa Clara, CA | 408-921-8963 | shash.v.mahesh@gmail.com

SKILLS

Languages:	Go, Java, Python, C/C++, SQL
Backend & Cloud:	AWS, Kafka, Kubernetes, Apigee, Docker, DynamoDB, MongoDB, PostgresSQL
Tools & Frameworks:	CUDA, PyTorch, Grafana, Prometheus, Elasticsearch, CI/CD, Git

WORK EXPERIENCE

DIRECTV – Backend Software Engineer — Jan 2023 – Present

- Led the design, implementation, and production rollout of a new **Unified Login System** serving **millions of users** across multiple auth methods spanning multiple microservices written in Go.
- Developed an **MCP Server** to enable AI agents to analyze Prometheus metrics & logs to aid root cause analysis.
- Module lead for new **authentication & authorization workflows** (manual login, SSO, 3rd-party, QR-code).
- Built a real-time fraud detection system that uses **Kafka** to ingest Auth events from multiple microservices and identifies suspicious login or authorization attempts using **ML (streaming-KMeans model)**.
- Revamped the user access **token generation, refresh,** & **revocation flow** to meet low-latency requirements, and removed reliance on third-party service.
- Built a tool to simulate AuthN/AuthZ SAML requests from third-party providers to internal services. Significantly sped up troubleshooting in production.
- Saved *$100,000* annually by migrating functionality from a legacy microservice to an **Apigee Proxy**.

DIRECTV – Software Dev & Architecture Intern — Jun 2022 – Sep 2022

- Saved over *$15 million* in annual revenue by building a new microservice with multiple advertising usecases (ad targeting & content recommendation) from ground up in the direct path of UX.
- Utilized an **internal cache** to reduce data lookups, to meet response latency targets.

NVIDIA – DriveIX Systems Software Intern — Aug 2021 – Dec 2021

- Used **CUDA** to blend and process camera and CV images to improve drive-view testing performance.
- Optimized security checks running on CPU resources with SIMD, improving efficiency by 50%.
- Assisted in building/debugging drivers to sensors; Resolved memory leaks & improved CPU efficiency.

Caterpillar – Digital Architecture & ML Intern — May 2021 – Aug 2021

- Trained ML models using AWS Sagemaker & automated daily retraining via AWS Step Function.
- Prepared data from Snowflake DB for ML use cases using AWS Data Wrangler.

EDUCATION

University of Illinois Urbana-Champaign

Bachelor of Science, Computer Engineering (Honors)	*GPA: 3.74/4.00*	Dec 2022
Master of Computer Science	*GPA: 4.00/4.00*	Aug 2024 – Present (Exp. 2026)

PROJECTS

- **RISC-V Processor**: Designed a 6-stage in-order pipelined RISC-V processor with caching; won class competition.
- **Linux-Based OS**: Built an OS using C/x86 Assembly with paging, file system, system calls, drivers, and scheduling.
- **MODIS Cloud Detection**: Developed ML model to detect clouds from MODIS satellite images, achieving 97.5% accuracy. Placed 2nd in NVIDIA NCSA Hackathon.